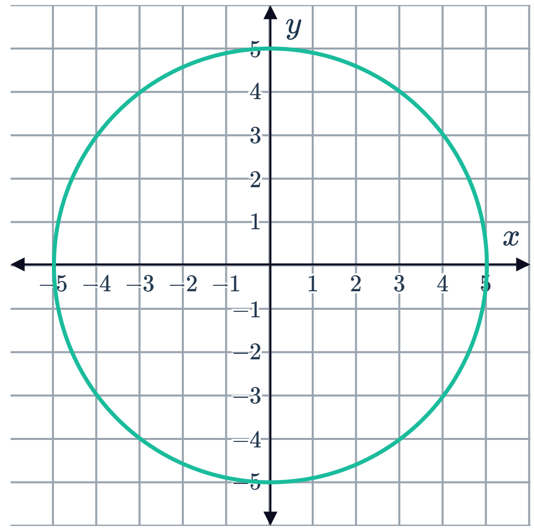
Graphs of circles

1 Consider the given graph of the circle.
Complete the statement:
Every point on the circle is exactly ▯ units away from the point (▯, ▯).



2 Determine whether the following points are inside, outside, or on the circle with equation $x^2 + y^2 = 25$:

a $(-3, 2)$ b $(4, 3)$ c $(1, 6)$ d $(-5, 0)$

3 For each of the following equations of circles:

i Find the radius. ii Sketch the graph of the circle.

a $x^2 + y^2 = 16$ b $x^2 + y^2 = 36$

c $x^2 + y^2 = 25$

4 Sketch the graph of the circle given by $x^2 + y^2 = 3$.

5 For each of the following equations:

i Find the center of the circle. ii Find the exact radius of the circle.

a $(x + 3.1)^2 + \left(y - \dfrac{3}{2}\right)^2 = \dfrac{9}{16}$ b $(x - 6)^2 + (y - 6)^2 = 12$

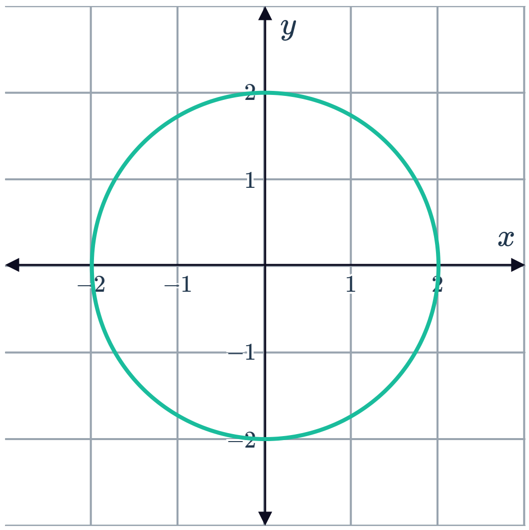
c $(x - 0.7)^2 + (y + 4.4)^2 = 3$ d $(x - 8)^2 + (y + 1)^2 = 13$

6 For each of the following circles, state:

i The coordinates of the center.

ii The radius.

a



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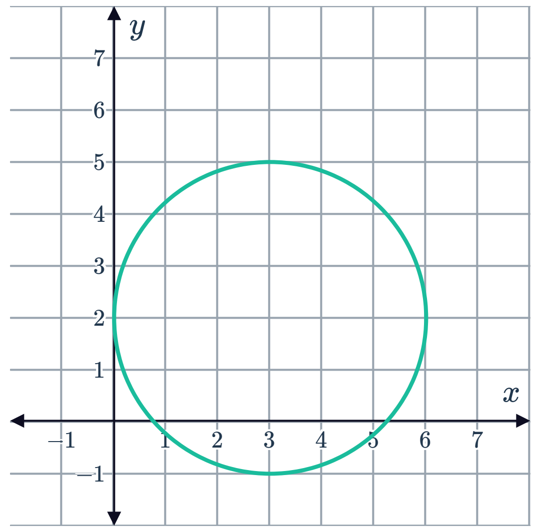

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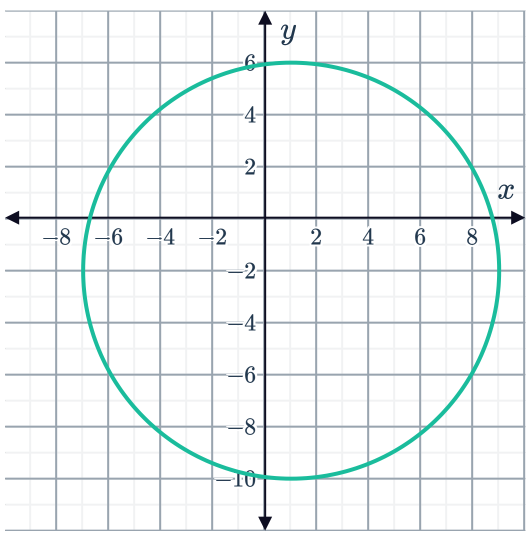

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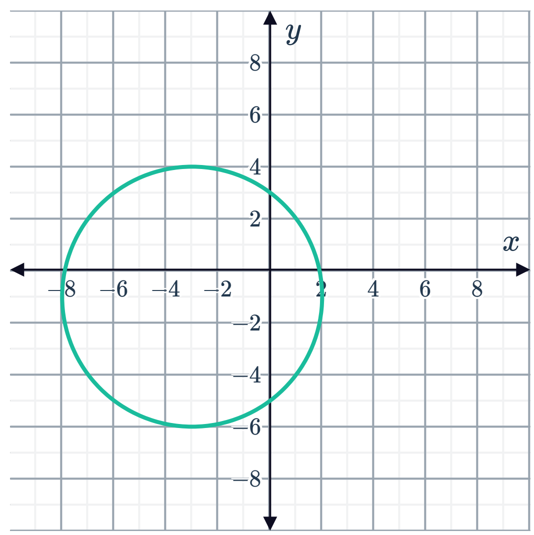


7 Sketch the graph of each of the following circles:

a $(x-2)^2 + y^2 = 9$

b $(x+3)^2 + y^2 = 16$

c $x^2 + (y+3)^2 = 6^2$

8 For each of the following equations:

i Find the center of the circle.

ii Find the radius of the circle.

iii Sketch the graph of the circle.

a $3x^2 + 3y^2 - 48 = 0$

b $25x^2 + 25y^2 = 400$

c $(x+3)^2 + (y+4)^2 = 16$

d $(x+4)^2 + (y-2)^2 = 25$

e $(x+1)^2 + (y+4)^2 = 25$

f $(x-4)^2 + (y+1)^2 = 36$

g $\left(x+\frac{1}{2}\right)^2 + \left(y+\frac{1}{2}\right)^2 = \frac{9}{4}$

9 Consider the point $(-40, -9)$ that lies on the circle centered at the origin with a radius of 40 units:

a Find the distance between $(-40, -9)$ and the origin.

b Does the point $(-40, -9)$ lie on the circle?

10 Consider a circle with center O at $(0, 0)$ that passes through the point $A(-7, 6)$. A diameter is drawn from point A to point B on the circle.
State the coordinates of point B.

11 Consider the points $A(1, -4)$ and $B(9, 2)$ and a circle with center $C(5, -1)$:

a Find the distance between points A and C.

b Find the distance between points B and C.

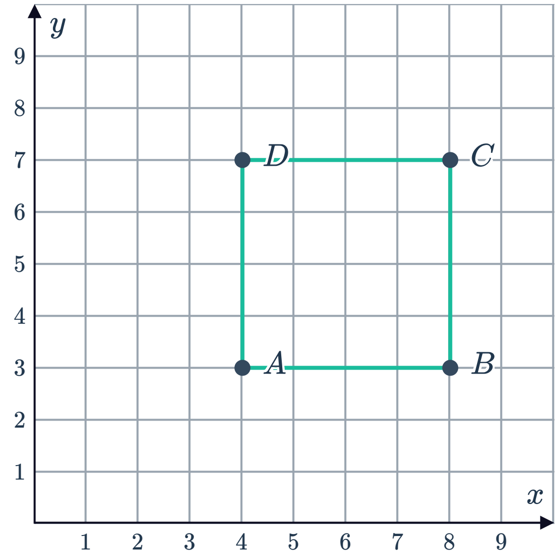
c State the coordinates of the midpoint of A and B.

d Do points A and B lie at opposite ends of the circle?

12 Consider a circle drawn to fit inside square $ABCD$:

a State the coordinates of the center of the circle.

b Find the radius of the circle.



Equations of circles

13 State whether the following are equations of circles:

a $(x+y)^2 = 4$ b $4x^2 + 5y^2 = 4$ c $y^2 = 2 + x^2$ d $4x^2 + 4y^2 = 16$

e $x^2 + y^2 = 4$ f $y^2 = 2 - x^2$

14 Consider the circle with center $(0,0)$ that passes through the point $(5,-10)$.

a Find the radius of the circle. b Find the equation of the circle.

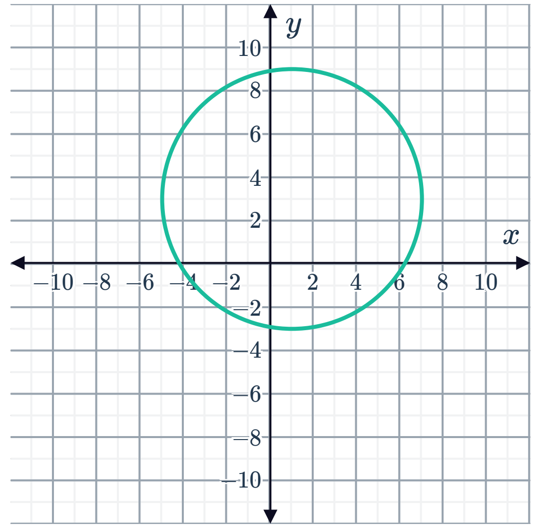
15 A circle with center $(0,0)$ has an x-intercept at $(6,0)$.

a Find the equation of the circle. b Find the exact area inside the circle.

16 Select the equation of the circle whose center is closer to the origin:

A $x^2 + (y+6)^2 = r^2$ **B** $x^2 + (y-2)^2 = r^2$

17 Consider the circle on the graph:



- **a** State the coordinates of the center of the circle.
- **b** Find the radius of the circle.
- **c** State the equation of the circle in standard form.

18 For each of the following circles:

- **i** Sketch the graph.
- **ii** Find the equation of the circle.

- **a** Circle with center at $(3,3)$ and a radius of 6 units.
- **b** Circle with center at $(0,6)$ and a radius of 2 units.
- **c** Circle with center at $(2,-6)$ and a radius of 3 units.
- **d** A circle with center at the origin and a radius of 4 units.
- **e** A circle with center at the origin and a radius of $\sqrt{3}$ units.
- **f** A circle has its center at the origin and a radius of $\sqrt{18}$ units.

19 Consider the circle with center $(-2,3)$ that passes through the point $(3,-4)$.

- **a** Find the radius of the circle.
- **b** State the equation of the circle in standard form.

20 For each of the following equations:

- **i** State the equation of the circle in standard form.
- **ii** State the coordinates of the center of the circle.
- **iii** Find the radius of the circle.

- **a** $x^2-8x+y^2-20=0$
- **b** $x^2+4x+y^2+6y-3=0$
- **c** $x^2-2x+y^2+8y-19=0$
- **d** $x^2+8x+y^2-6y+24=0$

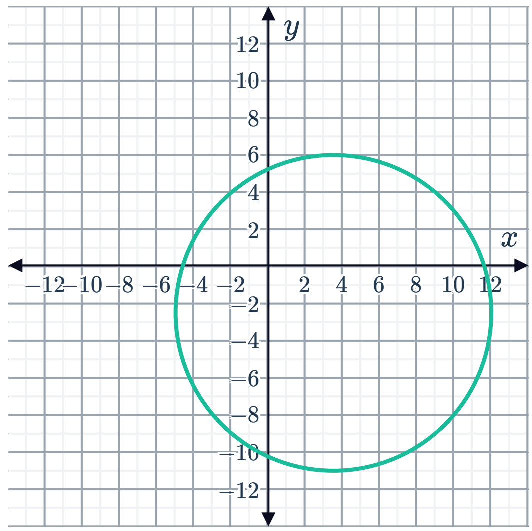
21 Consider the equation of a circle $x^2 + y^2 -$ [redacted] $y + \frac{27}{2} = 0$.

a State the equation of the circle in standard form.

b State the coordinates of the center of the circle.

c Find the radius of the circle.

d The circle that has been graphed has the same center as the circle with equation $x^2 + y^2 - 7x + 5y + \frac{27}{2} = 0$, but not the same radius.
Would the circle with equation $x^2 + y^2 - 7x + 5y + \frac{27}{2} = 0$ lie inside or outside the given circle?



22 For each of the following equations:

i State the equation of the circle in standard form .

ii State the coordinates of the center of the circle.

iii Find the radius of the circle.

iv Sketch the graph.

a $x^2 + y^2 + 2x + 4y = 11$

b $x^2 + 4x + y^2 - 10y = 52$

c $x^2 - 2x + y^2 - 16y - 16 = 0$

d $4x^2 + 4y^2 + 36x - 12y + 41 = 0$

23 Consider the equation of a circle given by $y^2 + 2y + 8 = 12x - x^2 + 7$.

a State the equation of the circle in standard form.

b State the coordinates of the center of this circle.

c Find the radius of this circle.

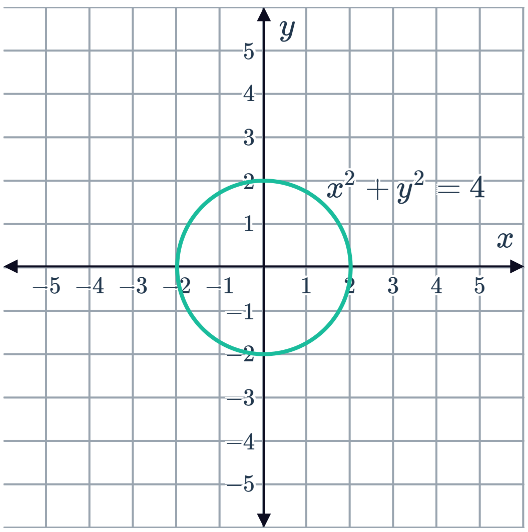
d Sketch the graph of this circle.

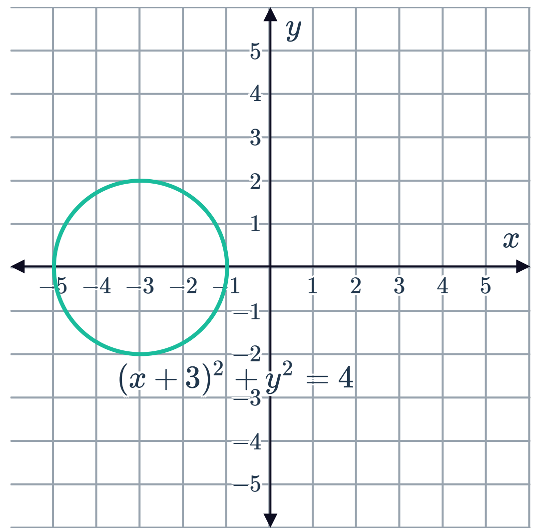
24 The endpoints of the diameter of a circle ar[redacted], $-3)$ and $(-15, -15)$.

- **a** State the coordinates of the center of the circle.
- **b** Find the exact radius of the circle.
- **c** State the equation of the circle which has $(-5, -3)$ and $(-15, -15)$ as endpoints of its diameter.

Translations of circles

25 Consider the following circles:





Describe the translation from $x^2 + y^2 = 4$ to $(x + 3)^2 + y^2 = 4$.

26 For each of the following translations, state the equation of the resulting circle:

- **a** A circle of radius 5, centerd at the origin, is translated 2 units downwards.
- **b** A circle of radius 7, centerd at the origin, is translated 3 units vertically upwards.
- **c** A circle of radius 3 units, centerd at the origin, is translated 4 units to the right.
- **d** A circle of radius 2 units, centerd at the origin, is translated 5 units to the left.

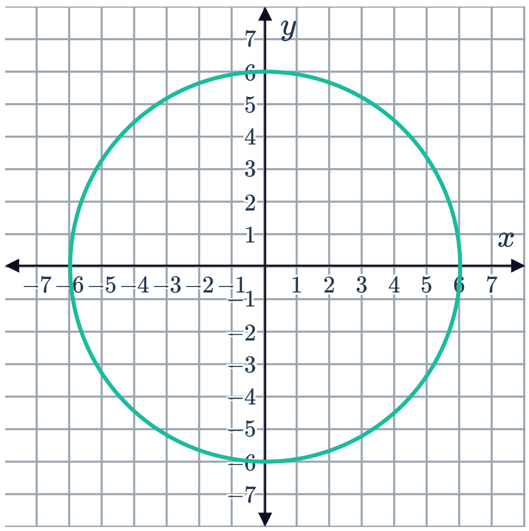
Domain and range of circles

27 For each of the following circles, state the following in interval notation:

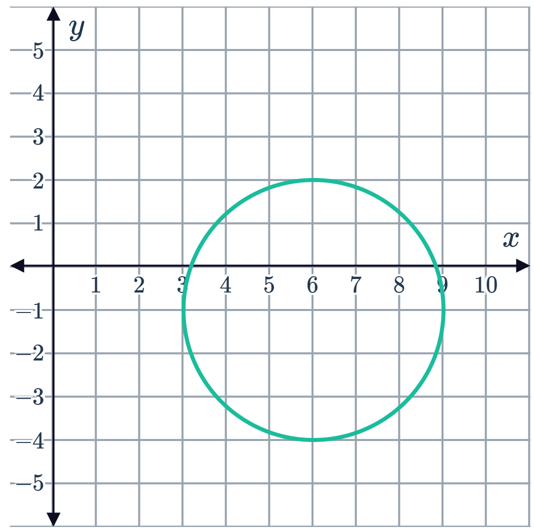
i The domain

ii The range

a



b



28 A circle has a radius of 1.5 and a center at $(-12,-13)$.

a State the domain of the graph in interval notation.

b State the range of the graph in interval notation.

c Is the point $(-13,-22)$ inside or outside of the circle?

29 For each of the following equations:

i Sketch the graph described by the equation.

ii State the domain of the graph in interval notation.

iii State the range of the graph in interval notation.

a $x^2+y^2=36$

b $x^2+(y-4)^2=4$

c $(x+5)^2+(y+3)^2=16$

30 A circle has a domain of $[2,10]$ and a range of $[-4,4]$.

a Sketch the graph of the circle.

b State the equation for the circle.

Applications

31 Naval ship A picks up the signal of another vessel 45 km away. Ship A can also see that there is a partner naval ship, ship B, 30 km south and 40 km west of it.

a Using $(0, 0)$ as the position of naval ship A, determine the distance between naval ships A and B.

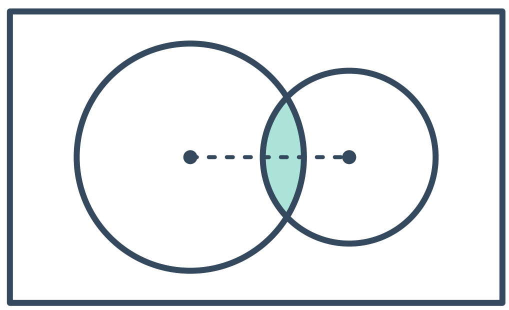
b Could the signal that was picked up be coming from naval ship B?

32 Three receiving stations are located on a coordinate plane, where each unit is one mile, at the points $(-3, -2)$, $(-1, 2)$ and $(3, 1)$. The epicenter of an earthquake is determined to be 2 miles, 4 miles and 5 miles from these three points respectively.

a Determine the coordinates of the epicenter of the earthquake.

b The earthquake can be felt 10 miles away. Determine if the town located at $(8, -7)$ would feel the earthquake.

33 Amy needs to use her school's laser cutter to make a cork component for her design project.
Using technology to sketch the component, she knows that the intersection of two different sized circles inscribed on a piece of cork gives the shape that she wants.



Two such circles have the equations $(x - 12)^2 + (y - 8)^2 = 49$ and $(x - 20)^2 + (y - 8)^2 = 36$, where each unit is 1 cm and $(0, 0)$ is the bottom left corner of the given piece of cork.

a State the domain and range of the leftmost circle in interval notation.

b State the domain and range of the rightmost circle in interval notation.

c What is the width of the component that Amy wants to make?

d Find the least area of material that could be used to make Amy's shape, considering the following constraints:

- The laser cutter will only accept rectangular pieces of material.
- The height of the intersection is 10.17 cm.
- There also needs to be a gap of at least 1 cm between any cut and the edge of the piece.

e If the bottom left corner of the rectangular piece in part (d) is now $(0, 0)$, state the new

34 A hamster wheel has a radius of $12 \mathrm{~cm}$. The clearance between the wheel and the ground is $5 \mathrm{~cm}$. The origin is directly below the center of the wheel. Use the coordinate system to write the equation of the circular hamster wheel.

35 A rectangular coordinate system with coordinates in kilometers is placed with the origin at the center of a city. Beth lives in a unit located $5.3 \mathrm{~km}$ west and $5.4 \mathrm{~km}$ south of the city center. She is researching malt shops in her area and finds that there is one estimated to be $30 \mathrm{~km}$ from her unit. Write the standard form of the equation for the set of points that could be the location of the malt shop.